sysMonitorManagement

Service Description

Service ID: *"sysMonitorManagement"*

Abstract
This document describes the `sysMonitorManagement` service, version 1.0.0, produced by the SysMonitor support system of Arrowhead 5.2.2.

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1 Overview

This document describes the `sysMonitorManagement` service, which provides selection of the sysMonitor observations SysMonitor collects: providers found by orchestration, subscriptions per provider and observation type. It is produced by the SysMonitor system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

1.1 Significant Prior Art

New in this workspace (SysMonitor-Management-HTTP-IDD-5.2.2).

1.2 How This Service Is Meant to Be Used

SysMonitor orchestrates all `sysMonitor` providers (generic_amqps, CERT_AUTH) every `ORCHESTRATION_INTERVAL_SECONDS` and binds their routing keys on the AMQP broker according to the subscriptions (default `all`). LocalCloudManager edits the subscriptions; `reconcile` re-orchestrates at once.

1.3 Important Delimitations and Dependencies

Only LocalCloudManager and Sysop. Subscription updates use optimistic locking (`If-Match` revision, 409 when stale).

2 Service Interface

This section describes the interfaces to the `sysMonitorManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

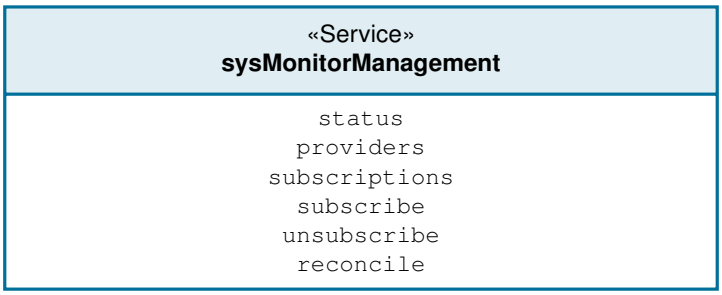


Figure 1: SysML-style block of the `sysMonitorManagement` service and its operations.

The following interface operations are available.

2.1 operation status () : Status

Returns the SysMonitor status.

2.2 operation providers () : ProviderList

Returns the orchestrated sysMonitor providers.

2.3 operation subscriptions () : SubscriptionList

Returns the subscriptions.

2.4 operation subscribe (SubscriptionRequest) : Subscription

Creates or updates the subscription for a provider and applies it to the broker bindings.

2.5 operation unsubscribe (String)

Removes the subscription for a provider (204).

2.6 operation reconcile () : ReconcileResult

Re-orchestrates the providers now (202).

3 Information Model

Here, all data objects that can be part of the `sysMonitorManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.7, which are used to represent things like hashes and identifiers.

3.1 struct ProviderList

Field	Type	Description
defaultSubscription	String	`all` or `none` (`SYSMONITOR_DEFAULT_SUBSCRIPTION`).
list	List<Map<String, Object>>	Orchestrated sysMonitor providers: `providerSystem`, `serviceInstanceId`, `baseRoutingKey`, `exchange`, `observationTypes`, `broker`, `port`, `orchestratedAt`.

3.2 struct ReconcileResult

Field	Type	Description
jobId	Identifier	Identifier of the started re-orchestration.

3.3 struct Status

Field	Type	Description
status	String	`running` or `degraded`.
system	String	`SysMonitor`.
implementation	String	`java`.
version	Version	`5.2.2`.
amqp	Map<String, Object>	Intake connection, bindings and last message time.
storage	Map<String, Object>	History counts, retention and pruning.
subscriptions	Number	Number of subscriptions.

3.4 struct Subscription

Field	Type	Description
providerSystem	String	Provider system.
observationTypes	List<String>	Selected types.
enabled	Boolean	Whether active.
revision	Number	Revision, also returned as `ETag`.
updatedAt	DateTime	Time of the last change.
updatedBy	Name	Last editor.

3.5 struct SubscriptionList

Field	Type	Description
defaultSubscription	String	`all` or `none`.
list	List<Subscription>	Subscriptions with `providerFound` and `unknownTypes`.

3.6 struct SubscriptionRequest

Field	Type	Description
observationTypes	List<String>	Types to receive; `*` for all (default).
enabled	Boolean	Default true.

3.7 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

4 References

[1] J. Delsing, “SysMonitor – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-SysMonitor-java-spring/documentation/v5.2.2/SysMonitor-v5.2.2-SysD.pdf.

[2] ——, “sysMonitorManagement HTTP/TLS/JSON – Interface Design Description (SysMonitor),” Eclipse Arrowhead, IDD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorManagement-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	